

Java I/O Assignment

1. **Objective:** Understand how to write data into a file using `PrintWriter`.

Task:

- Create a Java program that:
 - Takes 5 student details from the user: `Name`, `Age`, `Department`, `GPA`
 - Writes the details to a file `students.txt` using `PrintWriter`
 - Each line should be formatted as:
`Student[name=..., age=..., department=..., gpa=...]`

2. **Objective:** Learn how to format strings for cleaner output.

Task:

- Enhance the above program to:
 - Print each student detail in tabular format using `String.format()` with proper spacing.

Example:

Name	Age	Department	GPA
John	21	CSE	8.45
Priya	22	ECE	9.12

3. **Objective:** Learn to use parameterized string templates.

Task:

- Add functionality to:

- Generate a summary for each student using `MessageFormat`, like:
"Student {0} from {1} department has secured a GPA of {2}."
- Display this message on the console.

4. **Objective:** Understand object serialization.

Task:

- Create a `Student` class that implements `Serializable`
- Add properties: `id`, `name`, `age`, `department`, `gpa`
- After collecting 5 students:
 - Serialize them into a file called `students.ser`

5. **Objective:** Retrieve Java objects from a file.

Task:

- Read back the serialized data from `students.ser`
- Print the student objects in formatted output using `String.format` and `MessageFormat` again.